

Statistics Formula Sheet

Dentistry degree

Universidad CEU-Cardenal Herrera

Descriptive Statistics

Sample Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \dots + x_{n-1} + x_n}{n}$$

Sample Variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Standard Deviation

$$s = +\sqrt{s^2}$$

Percentile

100 p -th percentile is

$$(1 - \gamma) \cdot x_{(k)} + \gamma \cdot x_{(k+1)}$$

where:

k is the greatest integer less or equal than $(n - 1) \cdot p + 1$.

γ is the decimal part of $(n - 1) \cdot p + 1$, i.e. $\gamma = (n - 1) \cdot p + 1 - k$

Quartile

Q_1 =lower quartile = 25-th percentile.

Q_2 =median = 50-th percentile.

Q_3 =upper quartile = 75-th percentile.

InterQuartile Range

$$IQR = Q_3 - Q_1$$

Outlying Values

Smaller than $Q_1 - 1.5IQR$

Bigger than $Q_3 + 1.5IQR$

Confidence Intervals

Mean (large n)

$$(\bar{x} - z_{1-\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{s}{\sqrt{n}})$$

$$n > z_{1-\alpha/2}^2 \frac{\sigma^2}{\text{max.error}^2}$$

Mean (any n)

$$(\bar{x} - t_{n-1, 1-\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + t_{n-1, 1-\alpha/2} \frac{s}{\sqrt{n}})$$

Variance

$$((n-1) \frac{s^2}{\chi_{n-1, 1-\alpha/2}^2}, (n-1) \frac{s^2}{\chi_{n-1, \alpha/2}^2})$$

Proportions

$$(\hat{p} - z_{1-\alpha/2} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}, \hat{p} + z_{1-\alpha/2} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}})$$

$$n > \frac{z_{1-\alpha/2}^2 p_0(1-p_0)}{\text{max. error}^2}$$

One-sample Hyp. Testing

Mean (large n)

$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	p-value=2(1 - P(z < z)) (Two-sided HT)
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Mean (any n)

$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	Acceptance interval (Two-sided HT) (-t_{n-1, 1-\alpha/2}, t_{n-1, 1-\alpha/2})
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Proportion

$z = \frac{\hat{p} - p_0}{SE_{\hat{p}}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$
p-value=2(1 - P(Z < z)) (Two-sided HT)

Two-sample Hyp. Testing

Mean (with equal variances)

Pooled variance: $S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$
$t = \frac{(\bar{x}_1 - \bar{x}_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$
Acceptance interval (Two-sided HT) (-t_{n_1+n_2-2, 1-\alpha/2}, t_{n_1+n_2-2, 1-\alpha/2})

Mean (paired t -test)

$t = \frac{\bar{x}_D}{s_D/\sqrt{n}}$
Acceptance interval (Two-sided HT) (-t_{n-1, 1-\alpha/2}, t_{n-1, 1-\alpha/2})

Variance

$F_{d_1, d_2, \beta} = \frac{1}{F_{d_2, d_1, 1-\beta}}$ $F = \frac{s_1^2}{s_2^2}$
Acceptance interval (Two-sided HT) (F_{n_1-1, n_2-1, \alpha/2}, F_{n_1-1, n_2-1, 1-\alpha/2})

Two-sample Hypothesis Testing for Proportions

$\hat{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$
$z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$
p-value=2(1 - P(Z < z)) (Two-sided HT)

ANOVA

$$F = \frac{MSG}{MSE}$$

Acceptance Interval: [0, F_{k-1, n-k, 1-\alpha})

χ^2 -statistics

$$x^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Acceptance Interval: [0, $\chi_{(R-1) \cdot (C-1), 1-\alpha}^2$)